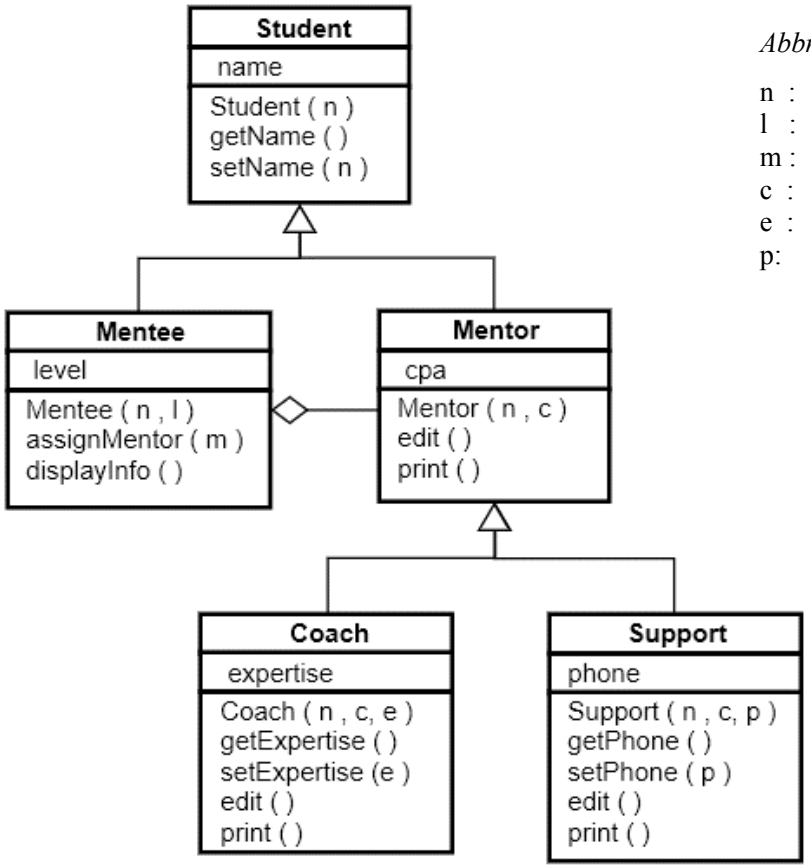
Program 4

School of Computing is planning to implement a mentoring programme involving all the students in the school. The programme will be run as follows:

- The students are divided into two categories, mentor and mentee. A mentor is a more knowledgeable student and offers helps to a mentee, a less knowledgeable student.
- Mentors are chosen from those students with good CPAs.
- Mentee's progress are kept track based on their levels, from level 1 to 5.
- There are two approaches of mentoring:
  - -Coaching-based: mentors will coach their mentees in particular areas, e.g., "C++ programming", "Game Programming", etc. Therefore, a coach must be expert in that area.
  - -Support-based: mentors offer their supports to the mentees without having to meet face to face. Thus, the mentors need to provide their phone numbers.



Abbreviations:

n : name
l : level
m : mentor
c : cpa
e : expertise
p: phone

Figure 4.1: Class diagram for Mentoring System

The school has hired you to develop a computer program to manage the records of the mentors and mentees. The class diagram for the program has been done by a senior software engineer and it is given in Figure 4.1 above. The classes are described further in Table 1 below.

Table 4.1: Description of the members of each class

Class Members	Description
class `Student`	
`name`	Student's name.
`Student(…)`	A constructor.
`getName()` and `setName(…)`	Accessor and mutator to the attribute.
class `Mentee`	
`level`	The class attribute indicating the level of progress for each mentee. There are five levels, from 1 to 5, where level 5 indicates mastery level.
`Mentee(…)`	A constructor.
`assignMentor(…)`	To assign a mentor to mentee.
`displayInfo()`	To display the mentee's information including: • name and level, • his/her mentor's details depending the type of the mentor, i.e. either a coach mentor or a support mentor. • If no mentor is assigned yet, then this method will print a message indicating that.
class `Mentor`	
`cpa`	Mentor's cpa.
`Mentor(…)`	A constructor.
`edit()`	To edit mentor information by the user.
`print()`	To print the mentor's name and cpa.
class `Coach`	
`expertise`	The mentor's area of expertise that he or she can be coaching, e.g., "C++ Programming", "Game Programming", etc.
`Coach(…)`	A constructor.
`getExpertise()` and `setExpertise(…)`	Accessor and mutator to the attribute.
`edit()`	To edit mentor's expertise by the user.
`print()`	To print the mentor's name, cpa and expertise.

class `Support`	
`phone`	The mentor's phone number.
`Support(…)`	A constructor.
`getPhone()` and `setPhone(…)`	Accessor and mutator to the attribute.
`edit()`	To edit mentor's phone number by the user.
`print()`	To print the mentor's name, cpa and phone.

Based on the class diagram in Figure 1 and specifications in Table 1, answer the questions (a) to (c) below. **Note:** As for questions 1 and 2, write your answers in a **separate text file**. Indicate the question numbers in your answers.

1. Determine the type of relationship formed between the following classes:

    a. `Mentee` and `Student`
    b. `Mentee` and `Mentor`
    c. `Coach` and `Mentor`
    d. `Coach` and `Student`
    e. `Coach` and `Mentee`

2. Analyse the class diagram in Figure 4.1 and its description in Table 4.1 as well as the program's output in Figure 4.2. Name TWO (2) methods that later in the the code, will be declared as virtual. Justify your answers.

3. Finally, implement the class diagram by modifying code in **exercise3.cpp**. Complete the following tasks in the program. Your code must follow the class diagram.

    a. Task 1. Delcare all the classes based on the class diagram.
    b. Task 2: Define all methods for the class `Mentor`
    c. Task 3: Define all methods for the class `Coach`
    d. Task 4: Define all methods for the class `Support`
    e. Task 5: Define all methods for the class `Mentee`
    f. Task 6: In the main function, an array of mentees and two mentors have been created. Complete the remaining tasks. See Task 6(a) to (h) in the program.

Output:

The program should produce the output as shown in Figure 2. Note that, **bold texts** indicate user inputs.

Screen 1: Displaying the current mentor.

```
========== Menu ==========
1. Select first mentor
2. Select second mentor
3. Display selected mentor
4. Edit selected mentor
5. Assign selected mentor to mentee
6. Unassign mentor from mentee
7. Display all mentees
9. Exit

Choose an operation [1-7, or 9] => 3

Current selected mentor: Ahmad Kamal
```

mentor1 is the current mentor

Screen 2: Assigning the current mentor to the first and second mentees (i.e index 0 and 1).

```
========== Menu ==========
1. Select first mentor
2. Select second mentor
3. Display selected mentor
4. Edit selected mentor
5. Assign selected mentor to mentee
6. Unassign mentor from mentee
7. Display all mentees
9. Exit

Choose an operation [1-7, or 9] => 5

Enter the index of mentee to assign with the current mentor => 0
```

<The menu is not included here for brevity>

```
Choose an operation [1-7, or 9] => 5

Enter the index of mentee to assign with the current mentor => 1
```

Screen 3: Displaying the mentee list.

<The menu is not included here for brevity>

```
Choose an operation [1-7, or 9] => 7

List of mentees:
#1
Mentee's  Name  : Abdul Samad
Mentee's  Level : 2
Mentoring type     : Coach-based
Mentor's Name      : Ahmad Kamal
Mentor's CPA       : 3.87
Mentor's Expertise : Programming

#2
Mentee's  Name  : Nurdiana
Mentee's  Level : 1
Mentoring type     : Coach-based
Mentor's Name      : Ahmad Kamal
Mentor's CPA       : 3.87
Mentor's Expertise : Programming

#3
Mentee's  Name  : Nurazlan
Mentee's  Level : 2
** No mentor yet **
```

The last mentee has yet to be assigned with a mentor

Screen 4: Changing the current mentor to another and assigning it to the last mentee (index 2).

```
========== Menu ==========
1. Select first mentor
2. Select second mentor
3. Display selected mentor
4. Edit selected mentor
5. Assign selected mentor to mentee
6. Unassign mentor from mentee
7. Display all mentees
9. Exit

Choose an operation [1-7, or 9] => 2

Second mentor is selected
```

<The menu is not included here for brevity>

```
Choose an operation [1-7, or 9] => 5

Enter the index of mentee to assign with the current mentor => 2
```

<The menu is not included here for brevity>

```
Choose an operation [1-7, or 9] => 7

List of mentees:
#1
Mentee's  Name  : Abdul Samad
Mentee's  Level : 2
Mentoring type     : Coach-based
Mentor's Name      : Ahmad Kamal
Mentor's CPA       : 3.87
Mentor's Expertise : Programming

#2
Mentee's  Name  : Nurdiana
Mentee's  Level : 1
Mentoring type     : Coach-based
Mentor's Name      : Ahmad Kamal
Mentor's CPA       : 3.87
Mentor's Expertise : Programming

#3
Mentee's  Name  : Nurazlan
Mentee's  Level : 2
Mentoring type : Support-based
Mentor's Name  : Siti Aminah
Mentor's CPA   : 3.98
Mentor's Phone : 0123456789
```

The last mentee now has a mentor

Screen 5: Changing the expertise info of the first mentor (i.e., `mentor1`) and the phone number for the second mentor (i.e. `mentor2`).

```
========== Menu ==========
1. Select first mentor
2. Select second mentor
3. Display selected mentor
4. Edit selected mentor
5. Assign selected mentor to mentee
6. Unassign mentor from mentee
7. Display all mentees
9. Exit

Choose an operation [1-7, or 9] => 1

First mentor is selected

<The menu is not included here for brevity>

Choose an operation [1-7, or 9] => 4

Mentor's Information:
Mentoring type     : Coach-based
Mentor's Name      : Ahmad Kamal
Mentor's CPA       : 3.87
Mentor's Expertise : Programming

You can only edit the mentor's expertise

Enter new expertise => Mobile and Web Programming

<The menu is not included here for brevity>

Choose an operation [1-7, or 9] => 2

Second mentor is selected

<The menu is not included here for brevity>

Choose an operation [1-7, or 9] => 4

Mentor's Information:
Mentoring type : Support-based
Mentor's Name  : Siti Aminah
Mentor's CPA   : 3.98
Mentor's Phone : 0123456789

You can only edit the mentor's phone number

Enter new phone number => +6013-89001000

<The menu is not included here for brevity>

Choose an operation [1-7, or 9] => 7
List of mentees:
```

Current mentor is mentor1 (a coach mentor). Thus, editing will be on coach data.

Current mentor is changed to mentor2 (a support-mentor). Thus, editing will be on support-mentor data.

```
#1
Mentee's  Name  : Abdul Samad
Mentee's  Level : 2
Mentoring type     : Coach-based
Mentor's Name      : Ahmad Kamal
Mentor's CPA       : 3.87
Mentor's Expertise : Mobile and Web Programming

#2
Mentee's  Name  : Nurdiana
Mentee's  Level : 1
Mentoring type     : Coach-based
Mentor's Name      : Ahmad Kamal
Mentor's CPA       : 3.87
Mentor's Expertise : Mobile and Web Programming

#3
Mentee's  Name  : Nurazlan
Mentee's  Level : 2
Mentoring type : Support-based
Mentor's Name  : Siti Aminah
Mentor's CPA   : 3.98
Mentor's Phone : +6013-89001000
```

Here, the mentor's expertise has changed accordingly.

Here, the mentor's expertise has changed accordingly.

Here, the mentor's phone number has changed accordingly.

Screen 6: Removing the mentor from the second mentee (i.e., at index 1).

```
<The menu is not included here for brevity>
Choose an operation [1-7, or 9] => 6

Enter the index of mentee to remove its mentor => 1

<The menu is not included here for brevity>
Choose an operation [1-7, or 9] => 7

List of mentees:
#1
Mentee's  Name  : Abdul Samad
Mentee's  Level : 2
Mentoring type     : Coach-based
Mentor's Name      : Ahmad Kamal
Mentor's CPA       : 3.87
Mentor's Expertise : Mobile and Web Programming

#2
Mentee's  Name  : Nurdiana
Mentee's  Level : 1
** No mentor yet **

#3
Mentee's  Name  : Nurazlan
Mentee's  Level : 2
Mentoring type : Support-based
Mentor's Name  : Siti Aminah
Mentor's CPA   : 3.98
Mentor's Phone : +6013-89001000
```

This mentee now does not have a mentor.

Figure 4.2: Program output